

Microsoft Copilot Studio and Power Virtual Agents

Microsoft Copilot Studio lets us create powerful AI-powered copilots for a range of requests, from providing simple answers to common questions to resolving issues requiring complex conversations. Engage with customers and employees in multiple languages across websites, mobile apps, Facebook, Microsoft Teams, or any channel supported by the Azure Bot Framework.

These copilots can be created easily without the need for data scientists or developers. Some of the ways that Copilot Studio copilots are used include:

- COVID-19 infection rate and tracking information.
- Sales help and support issues.
- Opening hours and store information.
- Employee health and vacation benefits.
- Common employee questions for businesses.

Copilot Studio is available as both a standalone web app, and as a discrete app within Teams. Most of the functionality between the two is the same. However, there might be different reasons to choose one version or the other based on the ways you want to use Copilot Studio.

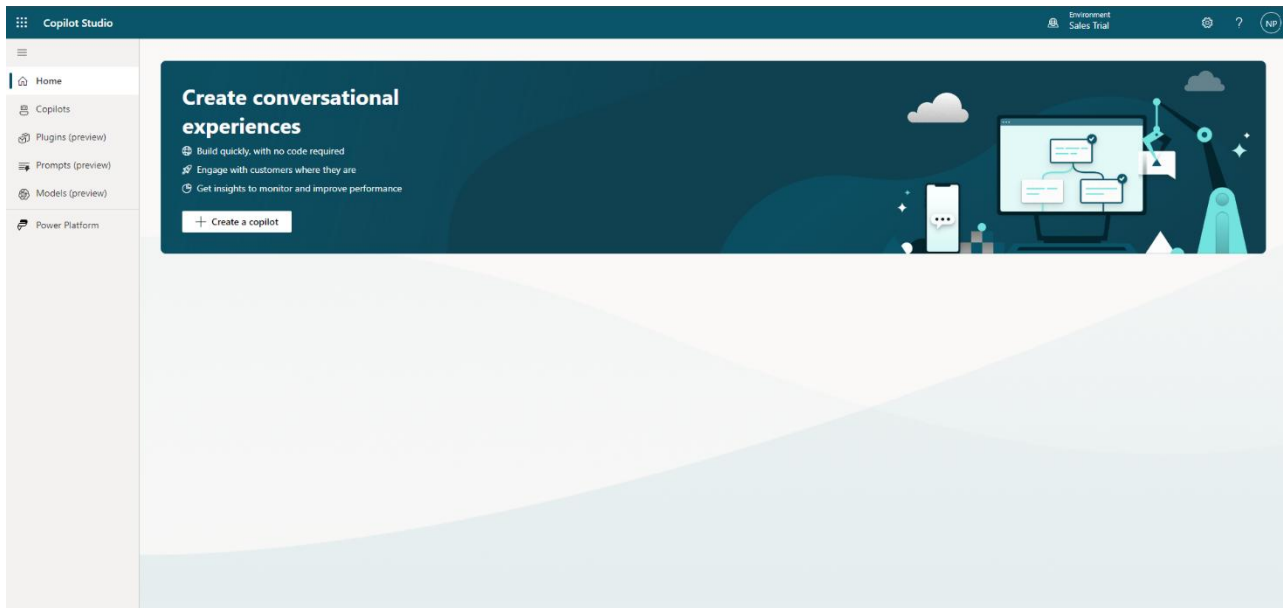
Copilot Studio version	Use cases
Web app at https://web.powervva.microsoft.com	- An IT admin who wants to create copilots for customers. - Familiar with chatbot services, you want to trial or test Copilot Studio. - A copilot user who wants to explore advanced copilot concepts, such as entities and variables, and create complex copilots.
Teams app	An employee or member of an organization who wants to use copilots to answer common employee questions. - You want to use advanced concepts, such as entities and variables, and have an internally-available copilot in Teams. - You want to create and distribute a copilot in the shortest time possible.

In common scenarios, there are situations where we are given a bot which communicate with us by means of predefined questions. For example, in e-commerce sites we have chatbots that provide the status of the order when we pass to them the unique identifier of the order. In that case, the chatbot simply accesses the CRM using the Guid and returns an existing information. Everything is predefined.

Navigate to <https://learn.microsoft.com/en-us/microsoft-copilot-studio/fundamentals-get-started?tabs=web> for a basic guide on how to create a deploy a Microsoft Copilot Studio copilot.

1. Create a chatbot

Let us navigate to <https://web.powerva.microsoft.com>. The following is the home page of Copilot Studio.



Click on + Create a copilot.

The screenshot shows the "Create a copilot" form. On the left, under "Set up the copilot", it says "Start fresh with a new copilot, and start making it yours." The main form area contains: a "Copilot name" field with the text "Expense Power Pages Bot"; a "What language do you want your copilot to speak?" dropdown menu set to "English (en-US)"; a section titled "Give your copilot some knowledge by setting up your Generative AI" with a sub-header "Add knowledge to your copilot by pointing it to an external website so your copilot can instantly answer questions over your data. Learn more"; and an "Enter your website" text input field. At the bottom, there are three buttons: "Edit advanced options", "Create", and "Cancel".

Copilot Studio

Environment Sales Trial

?

NP

Create a copilot

Set up the copilot

Start fresh with a new copilot, and start making it yours.

Copilot icon *

Lesson topics *

☒ Include lesson topics, which help users understand how topics work

Advanced options

Solution *

Demo

Schema name *

demo_expensePowerPagesBot

Voice Capabilities

Want to enable voice capabilities for your copilot? [Start here](#)

< Back

Create

Cancel

Finally, click on Create.

Copilot Studio | Expense Power Pages Bot

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Copilots

Overview

Generative AI

Topics & Plugins

Entities

Analytics

Publish

Extend Microsoft Copilot (preview)

Settings

Test copilot

Tracking: off

Chat

Hello, I'm Expense Power Pages Bot, a virtual assistant. Just so you are aware, I sometimes use AI to answer your questions. If you provided a website during creation, try asking me about it! Need try giving me some more knowledge by setting up generative AI.

just now

Type your message

Expense Power Pages Bot

NP

Share

Set up your generative AI

Add knowledge to your copilot by adding an external website, a SharePoint site, uploading files, and more so your copilot can instantly answer questions over your data. [Learn more](#)

Add a website

Allow your copilot to chat over the content of websites you choose.

Enter your website

Add a website

Add files

Allow your copilot to chat over the content of files you upload.

Add files

Manage generative AI

Manage your Generative AI settings, data sources, and enable even more powerful capabilities.

Go to Generative AI

Create custom topics

Build custom topics and workflows for your copilot to use.

Create topic

Create Plugins & Actions

Connect to your enterprise data, call APIs, connectors, flows and more - and expose them as plugins for your copilot to use.

Create plugin action

Extend a Microsoft Copilot (preview)

Tailor your Microsoft Copilot for your enterprise needs with plugins.

Extend with conversational plugins

Learn more

Tell us what you think

To begin, we start by clicking on Topics & Plugins.

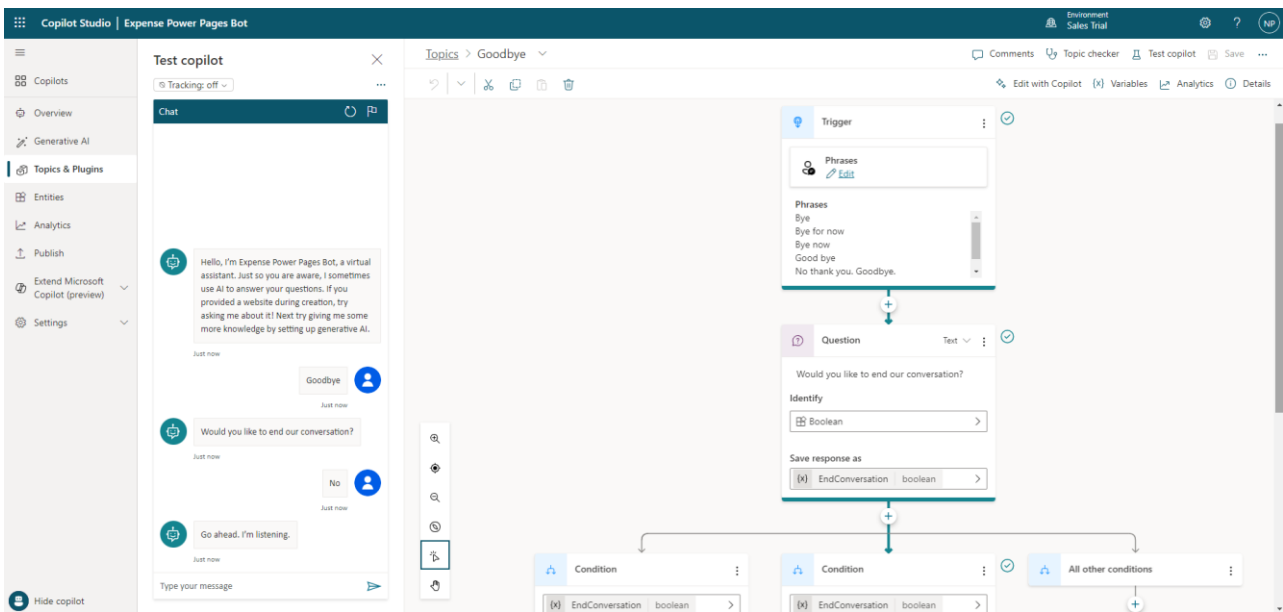
Name	Trigger	Description	Errors	Editing	Modified by
Goodbye	Phrases	This topic triggers when the user says goodbye. By...			NP Nicola Paracone 5 hours ago
Greeting	Phrases	This topic is triggered when the user greets the bot.			NP Nicola Paracone 5 hours ago
Lesson 1 - A simple topic	Phrases				NP Nicola Paracone 5 hours ago
Lesson 2 - A simple topic with a condition and vari...	Phrases				NP Nicola Paracone 5 hours ago
Lesson 3 - A topic with a condition, variables and a...	Phrases				NP Nicola Paracone 5 hours ago
Start Over	Phrases				NP Nicola Paracone 5 hours ago
Thank you	Phrases	This topic triggers when the user says thank you.			NP Nicola Paracone 5 hours ago

Here we have a collection of predefined topics which already allow us to interact with the chatbot.

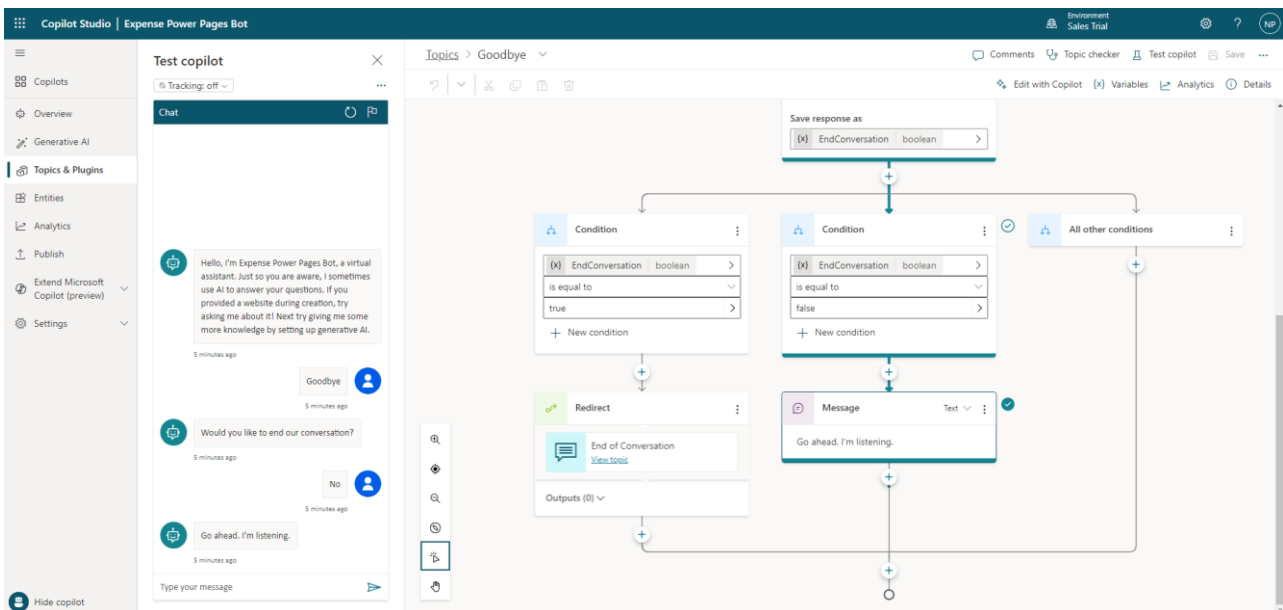
Name	Trigger	Description	Errors	Editing	Modified by
Goodbye	Phrases	This topic triggers when the user says goodbye. By...			NP Nicola Paracone 5 hours ago
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Lesson 1 - A simple topic	Phrases				NP Nicola Paracone 5 hours ago
Lesson 2 - A simple topic with a condition and vari...	Phrases				NP Nicola Paracone 5 hours ago
Lesson 3 - A topic with a condition, variables and a...	Phrases				NP Nicola Paracone 5 hours ago
Start Over	Phrases				NP Nicola Paracone 5 hours ago
Thank you	Phrases	This topic triggers when the user says thank you.			NP Nicola Paracone 5 hours ago

If we type "Goodbye", the chatbot already knows how to answer.

The instructions it follows are contained in the related topic, called Goodbye.



When the user types phrases such as "Bye", "Bye for now", "Good bye", etc., the chatbot is triggered. It then asks to the user "Would you like to end our conversation?" and stores the answer as a Boolean.



If the answer evaluates to false, the chatbot writes "Go ahead. I'm listening." and restarts the loop, otherwise it closes the conversation and redirects the user to the topic called End of Conversation. This will have a logical structure based on questions and answers from the user.

Let us create a new topic by click on + Add.

The screenshot shows the Copilot Studio interface for 'Expense Power Pages Bot'. The left sidebar contains navigation options: Copilots, Overview, Generative AI, Topics & Plugins, Entities, Analytics, Publish, Extend Microsoft Copilot (preview), and Settings. The 'Topics & Plugins' section is active, displaying a table of existing topics. A '+ Add' button is visible at the top left of the table, with a dropdown menu showing options: 'Topic', 'Plugin action (preview)', 'Microsoft Copilot plugin (preview)', 'From blank', and 'Create from description'. The table lists topics such as 'Goodbye', 'Greeting', 'Lesson 1 - A simple topic', 'Lesson 2 - A simple topic with a condition and variable', 'Lesson 3 - A topic with a condition, variables and a pre-built entity', 'Start Over', and 'Thank you'. Each row includes columns for Name, Trigger, Description, Errors, Editing, and Modified by.

Name	Trigger	Description	Errors	Editing	Modified by
Goodbye	Phrases	This topic triggers when the user says goodbye. By default, it does not end...			NP Nicola Paracone 5 hours ago
Greeting	Phrases	This topic is triggered when the user greets the bot.			NP Nicola Paracone 5 hours ago
Lesson 1 - A simple topic	Phrases				NP Nicola Paracone 5 hours ago
Lesson 2 - A simple topic with a condition and variable	Phrases				NP Nicola Paracone 5 hours ago
Lesson 3 - A topic with a condition, variables and a pre-built entity	Phrases				NP Nicola Paracone 5 hours ago
Start Over	Phrases				NP Nicola Paracone 5 hours ago
Thank you	Phrases	This topic triggers when the user says thank you.			NP Nicola Paracone 5 hours ago

The screenshot shows the 'Topics > Untitled' view in Copilot Studio. The left sidebar is the same as the previous image. The main area displays a trigger configuration panel for a new topic. The panel has a 'Trigger' section with a 'Phrases' input field and an 'Edit' link. Below this, there is a 'Phrases' section with the text 'No phrases to show'. To the right of the main area, there is a 'On Recognized Intent properties' panel. This panel includes a 'Condition' section with a 'Builder' dropdown, a 'Select a variable' dropdown, and a 'Condition' dropdown. It also has a 'Priority' section with a description and an 'Enter number' input field.

On Recognized Intent properties

Triggered when an intent is recognized.

Phrases

Phrases teach the copilot different ways someone might ask about this topic.

Condition Builder

Select a variable

Condition

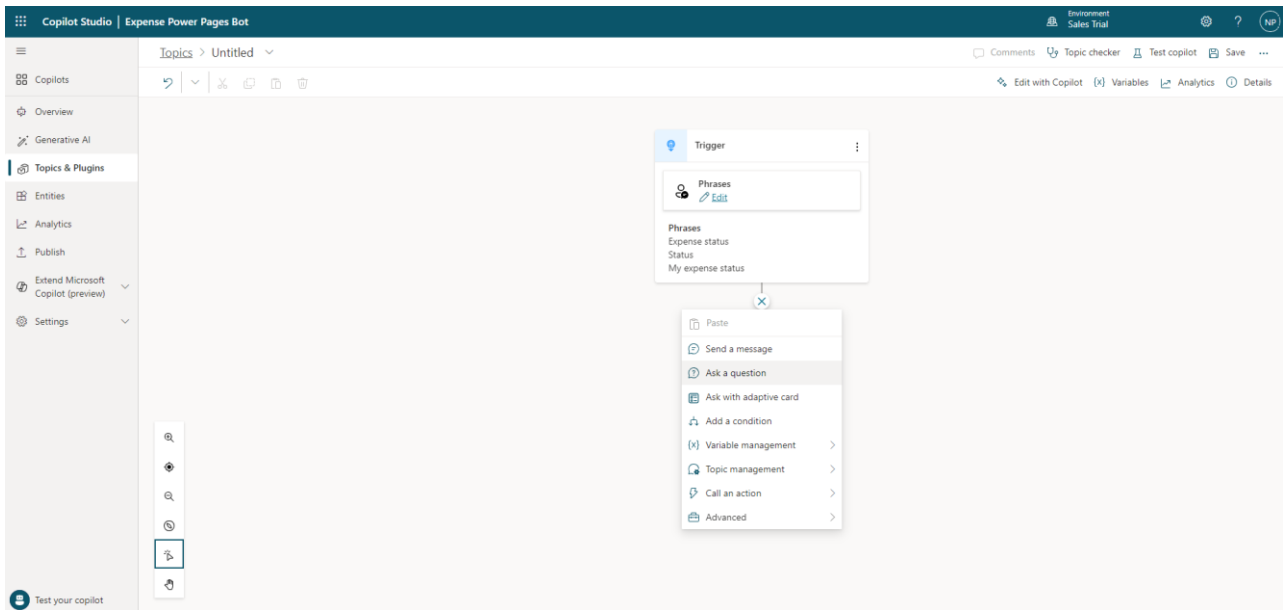
Enter or select a value

Priority

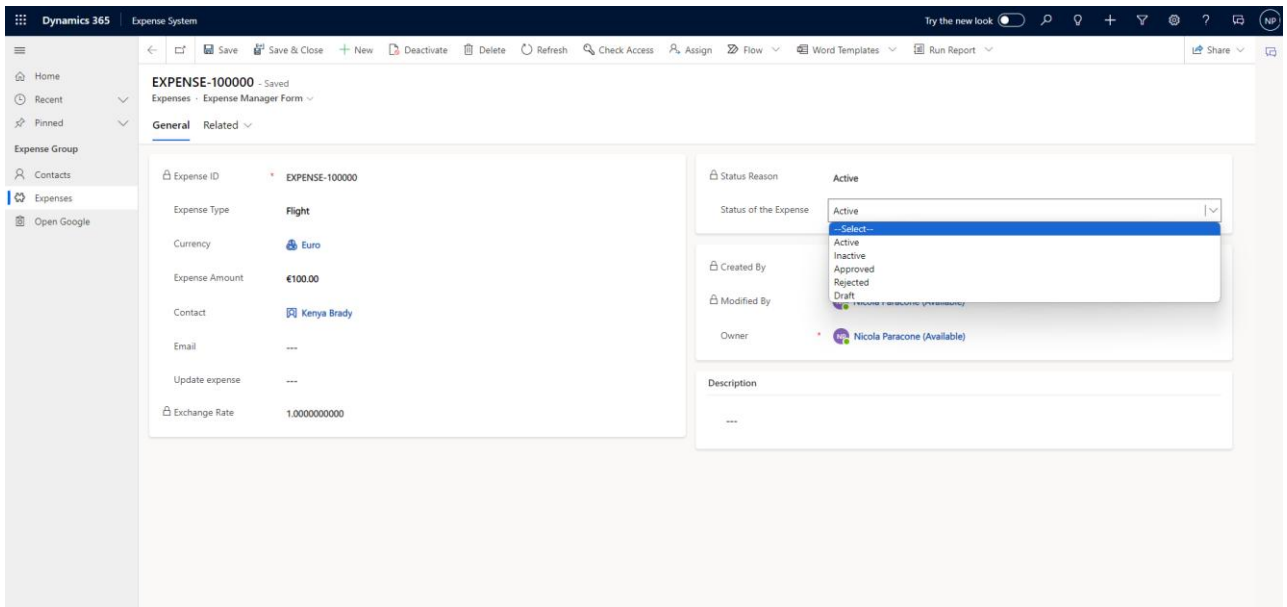
Set priority for this trigger against other qualifying triggers with lower numbers indicating higher priority.

Enter number

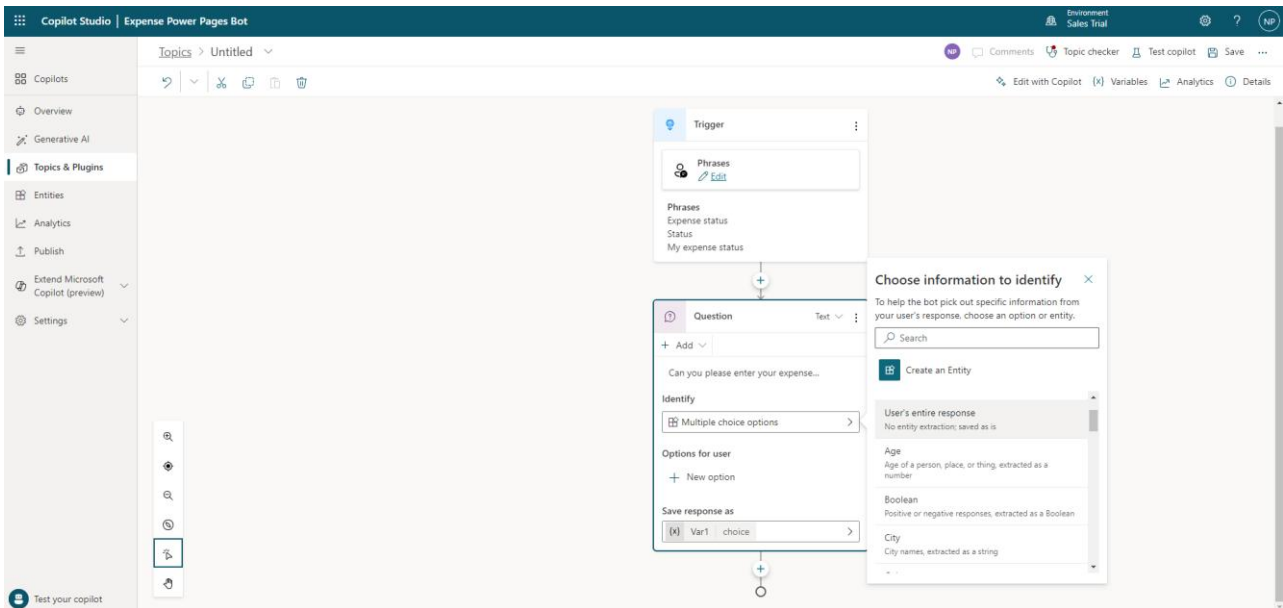
In relation to our Expense System application in Dynamics 365, we define some phrases that a user can type to trigger the chatbot. Then we want the chatbot to ask a question.



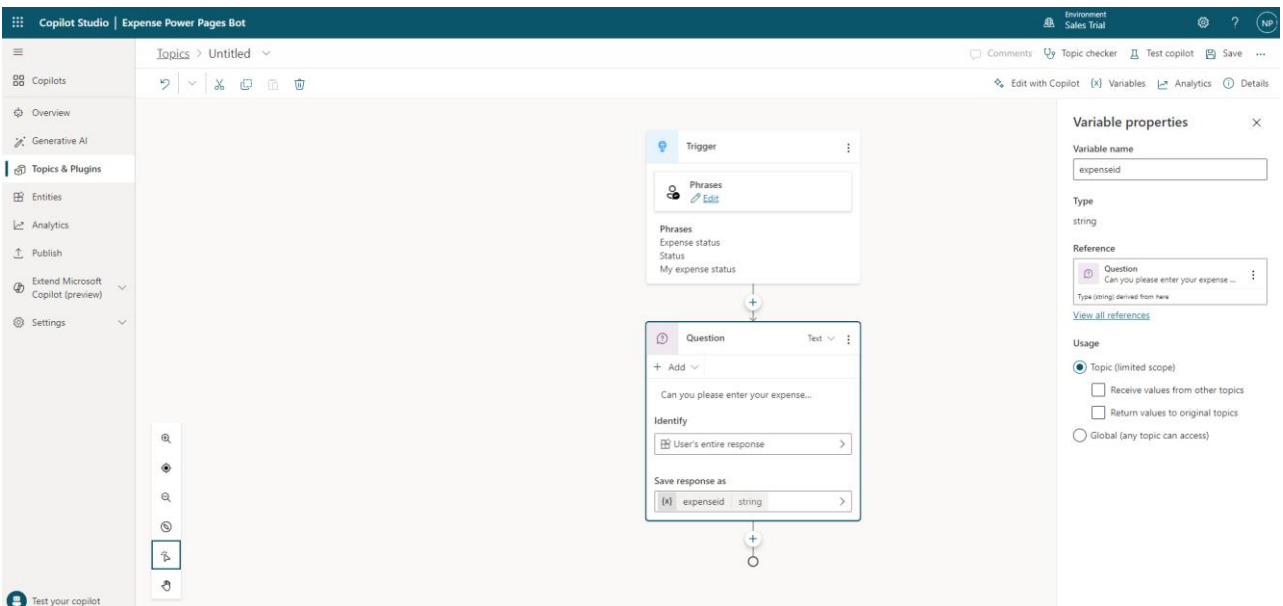
For the purpose of this demo, we have edited the Expense Manager Form as follows. Besides some refactoring of the interface, we have added the field Status of the Expense, which shows if the expense has been accepted, rejected, and so on.



The chatbot asks the user to input the Id associated with the expense they want to retrieve information of. Such input is saved as it is.

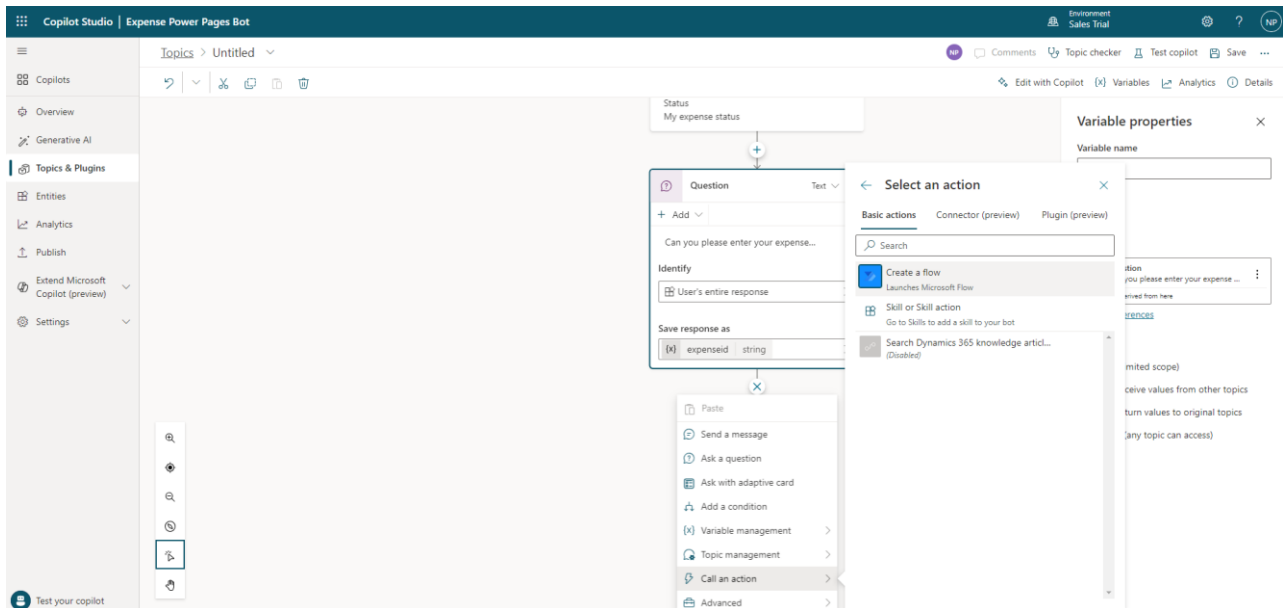


We store it into a string variable called expenseid.

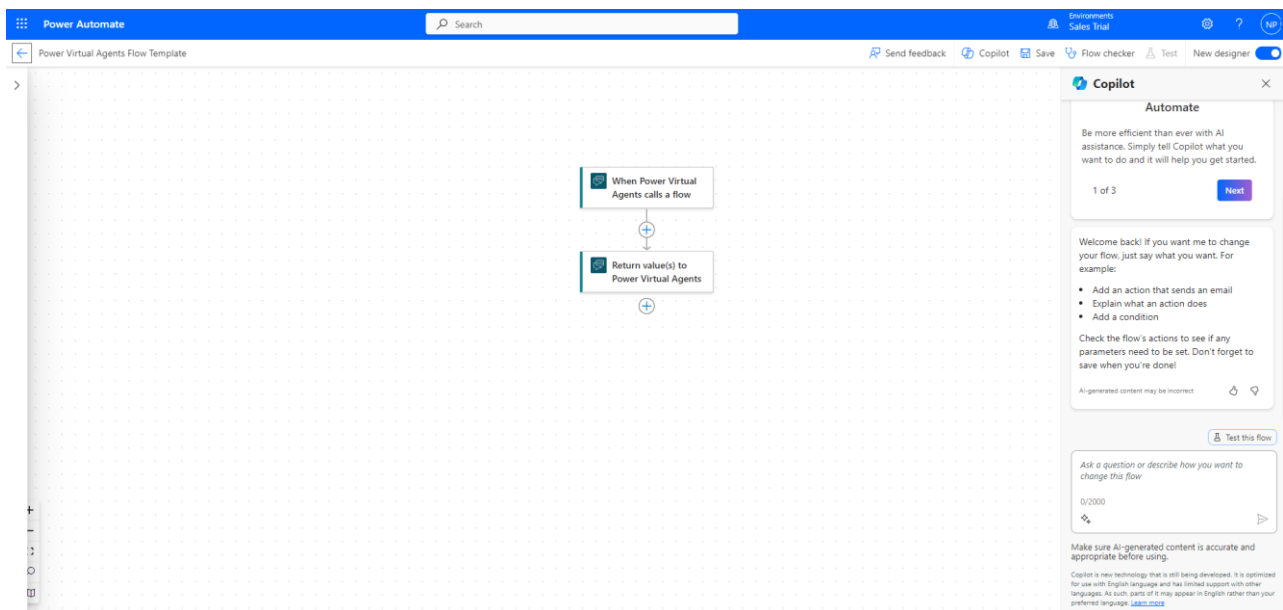


At this point, the intent is to retrieve the status of the expense associated with that Id.

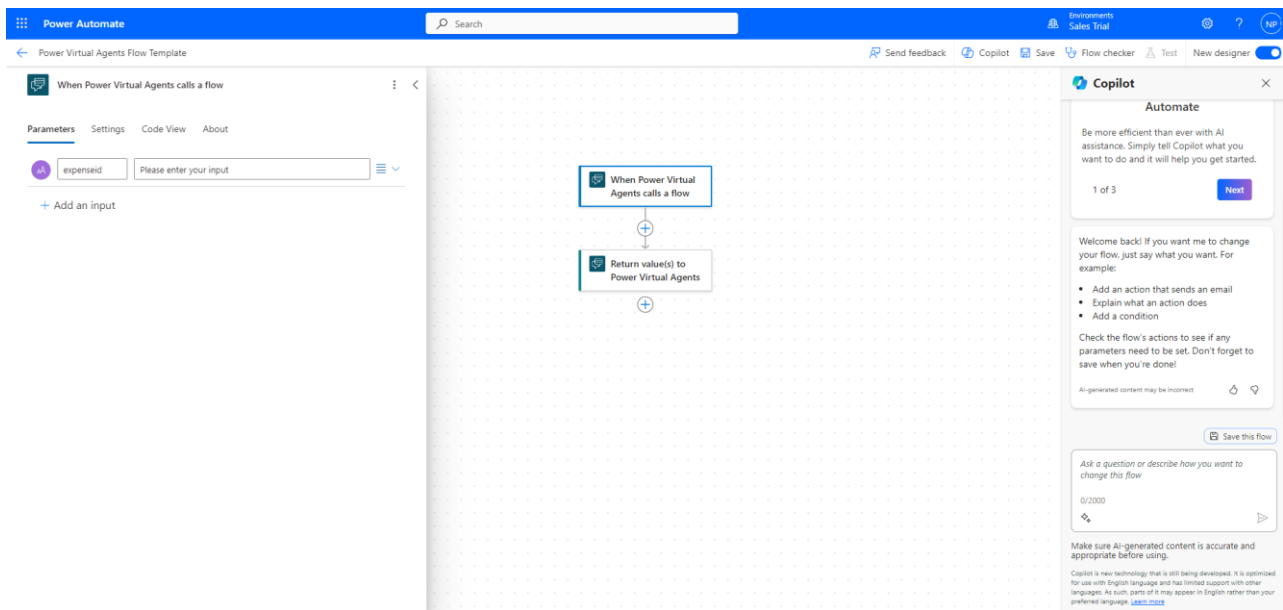
To do that, in the subsequent step we use a Call an action to launch a flow which retrieves the expense we are interested in.



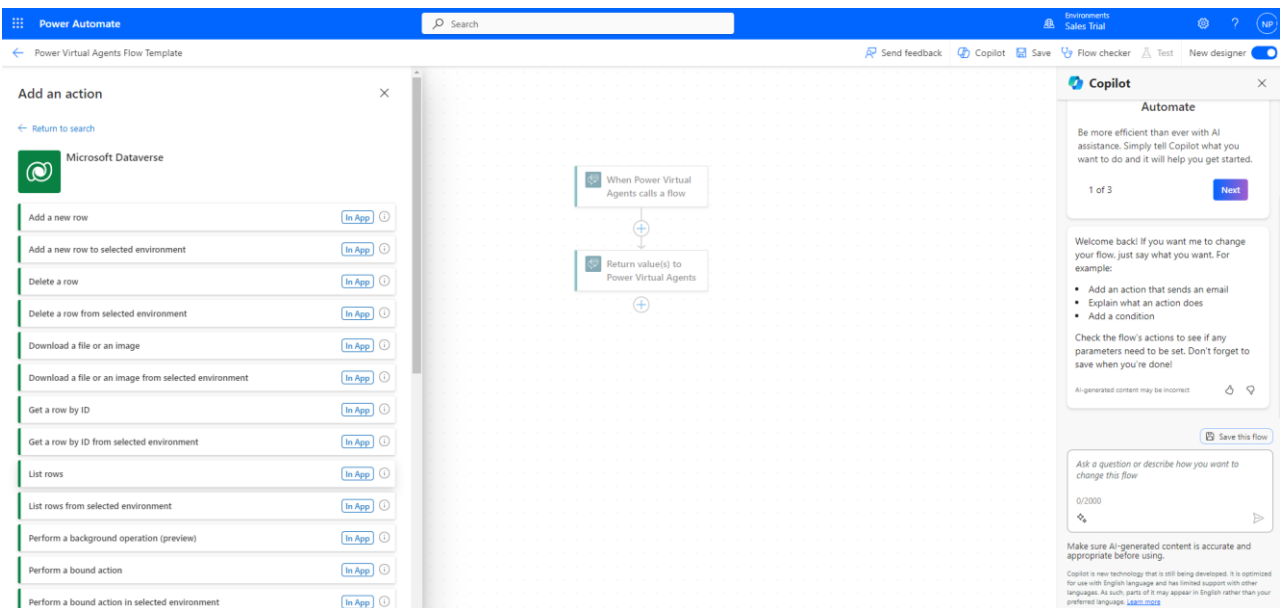
Doing so, we are automatically redirected to Power Automate.

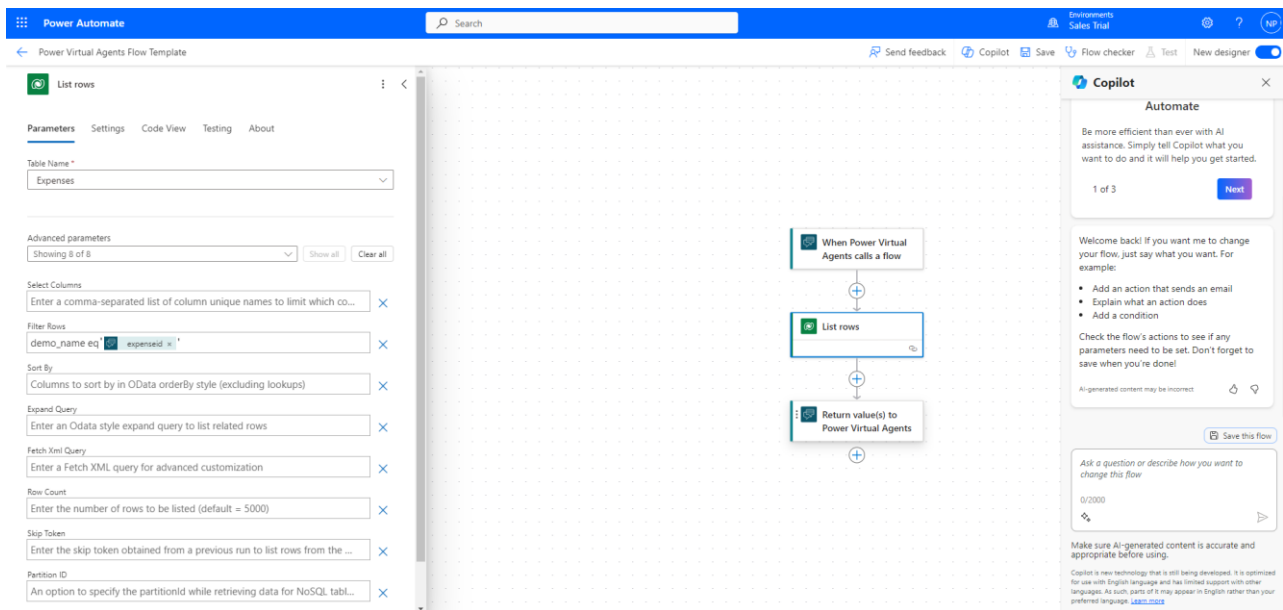


First, we have to provide an input to trigger, namely the expenseid variable.

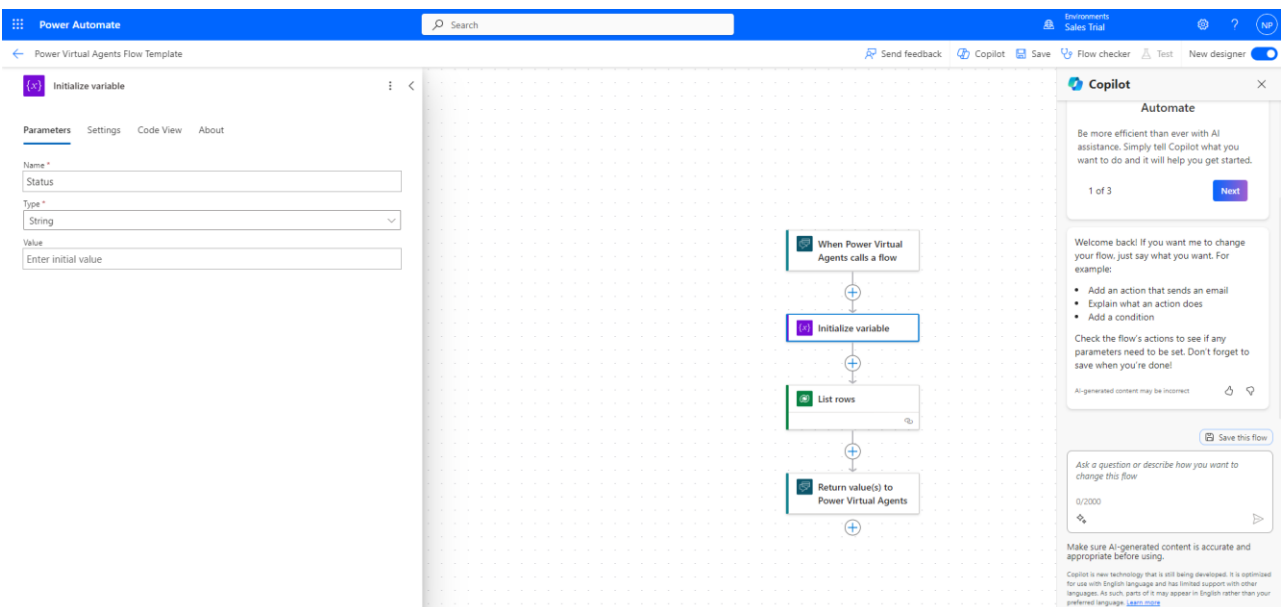


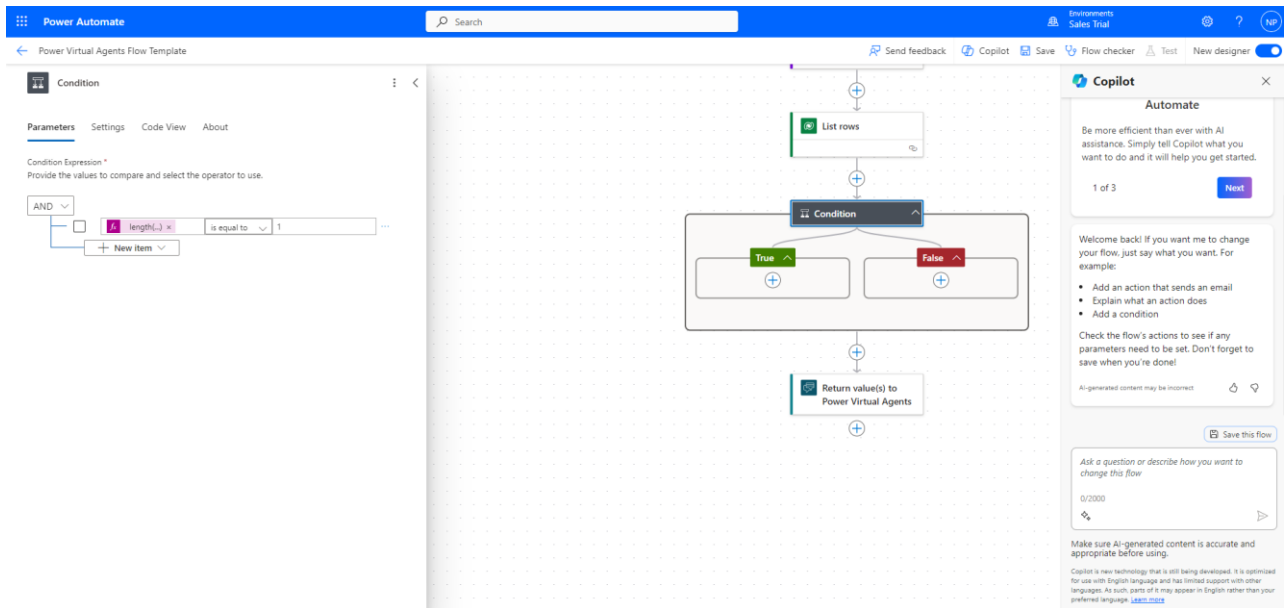
Using that expenseid, we want to connect to our Dataverse and retrieve the record. To do that, we add a List rows action belonging to the Microsoft Dataverse connector.



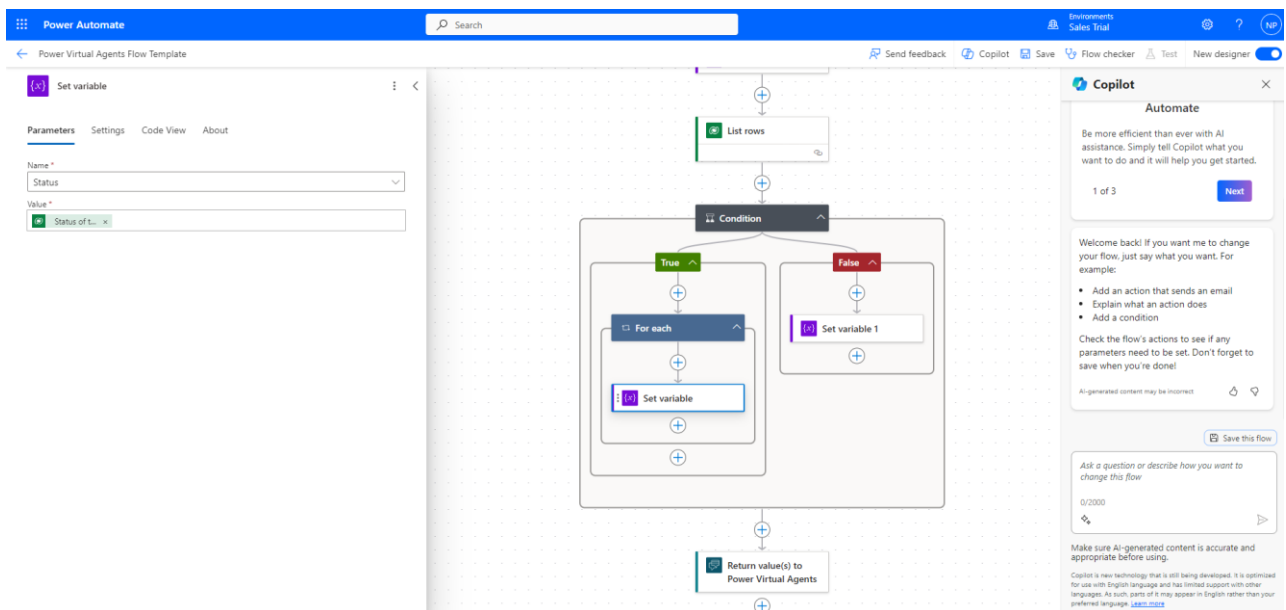


We also initialize a string variable, which will contain the status of the retrieved expense.

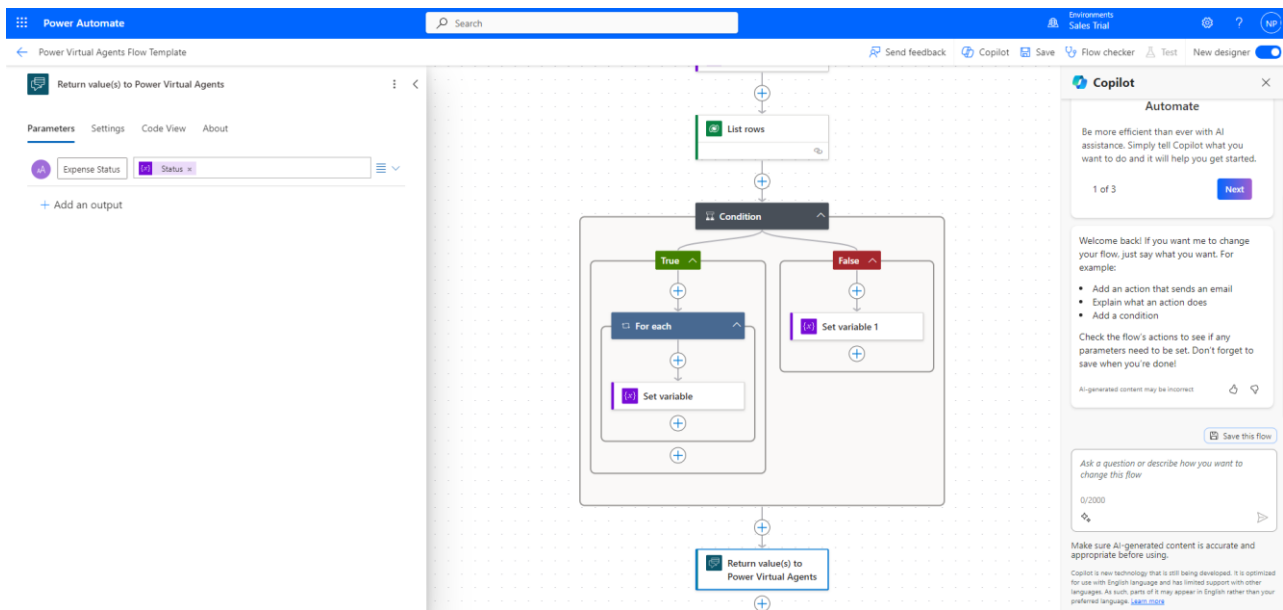




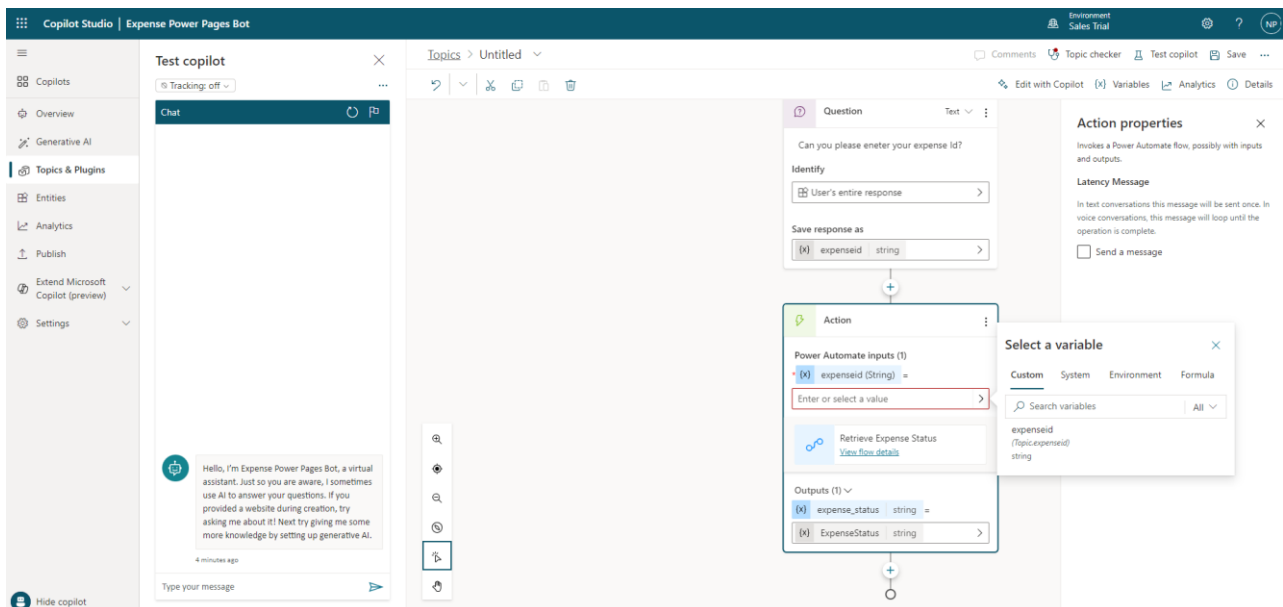
If the length of the list of retrieved expenses is equal to one, we set our variable to the value of the Status of the Expense field. Otherwise, we will set it to "Invalid expense Id".



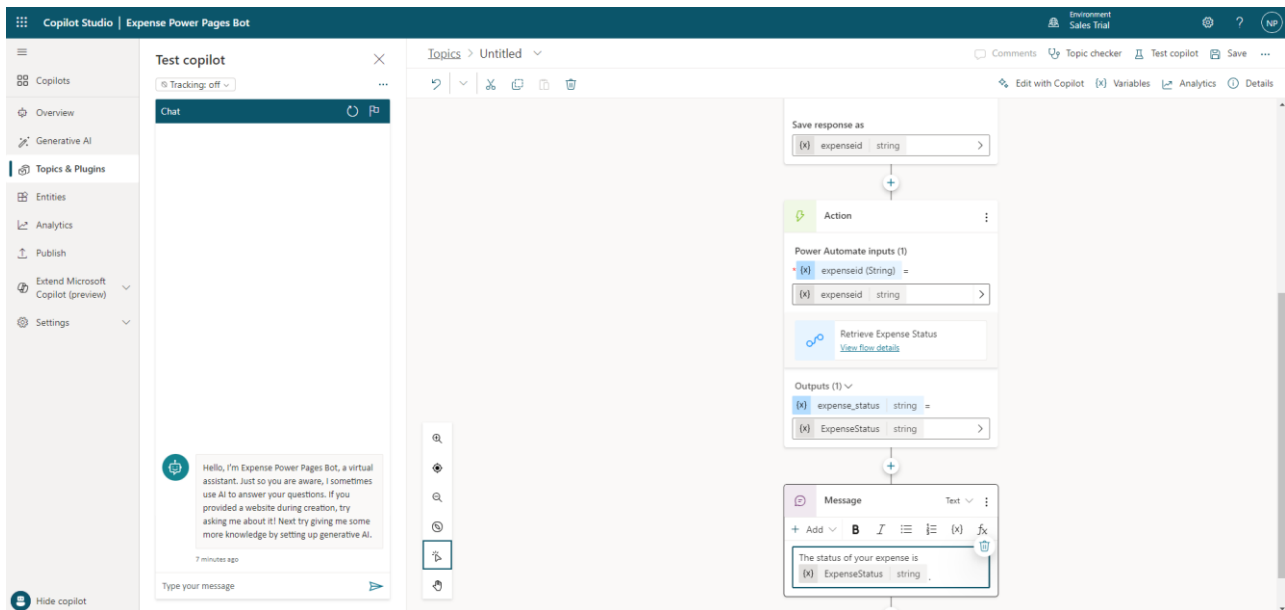
Finally, the value which will be returned to Power Virtual Agents is that of Expense Status.



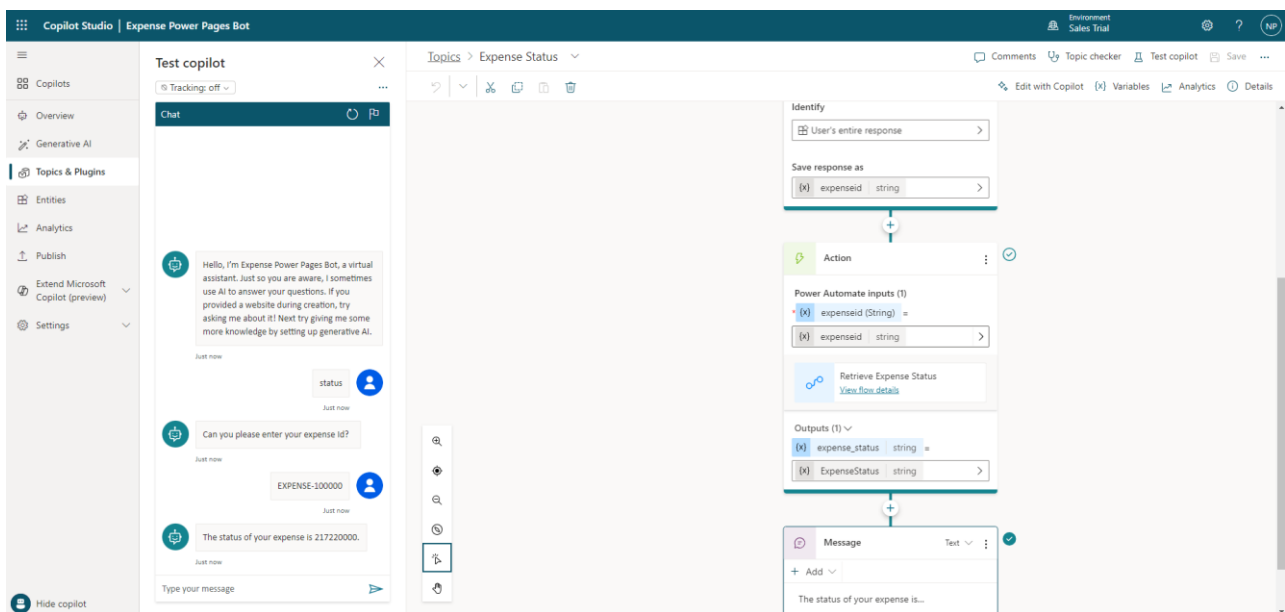
Let's save the flow with the name Retrieve Expense Status.



Now that the chatbot has retrieved the status of that particular expense, all it has to do is show it to the user.

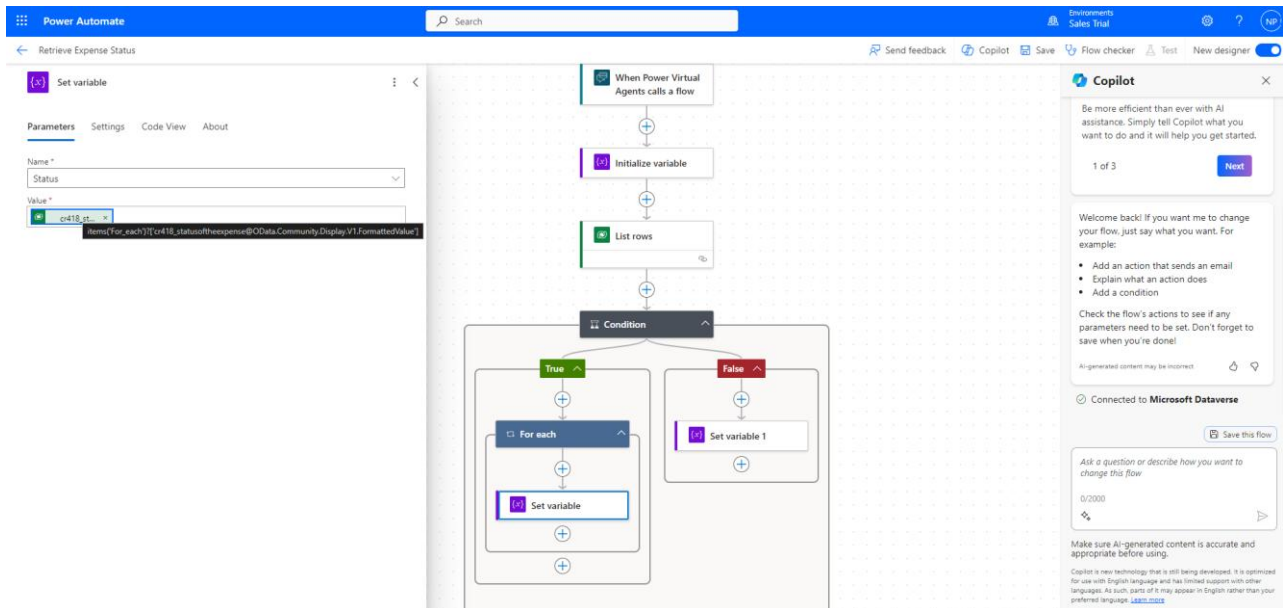


Finally, we can save the topic with name Expense Status and test the bot.

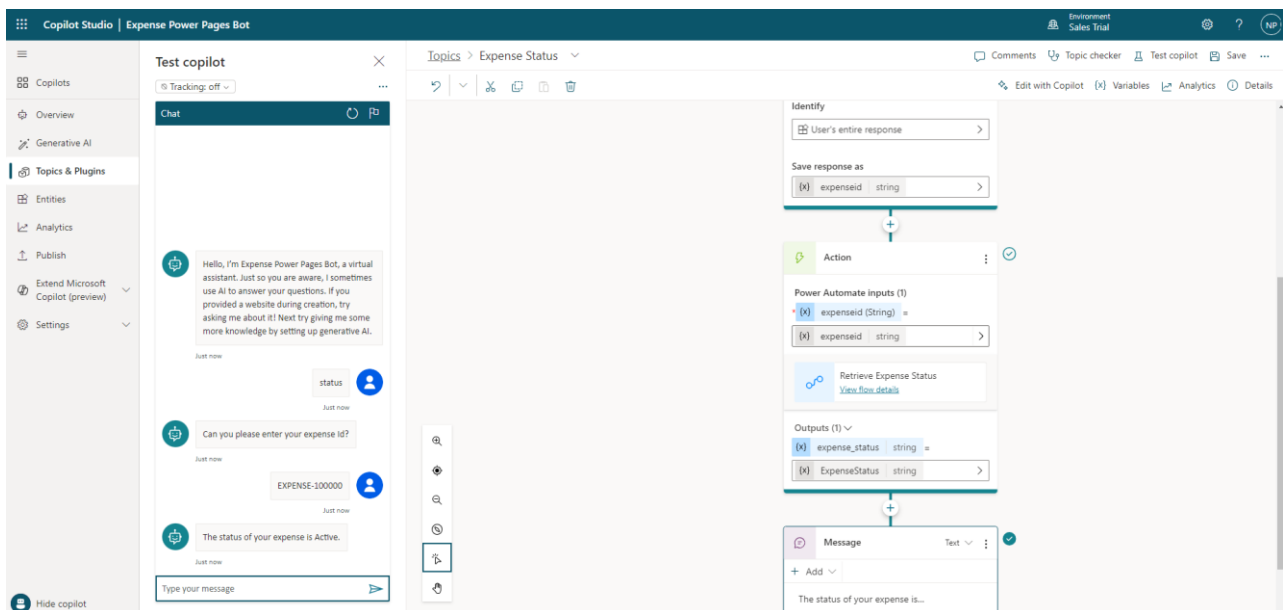


Notice that, by default, the bot returns the optionset value.

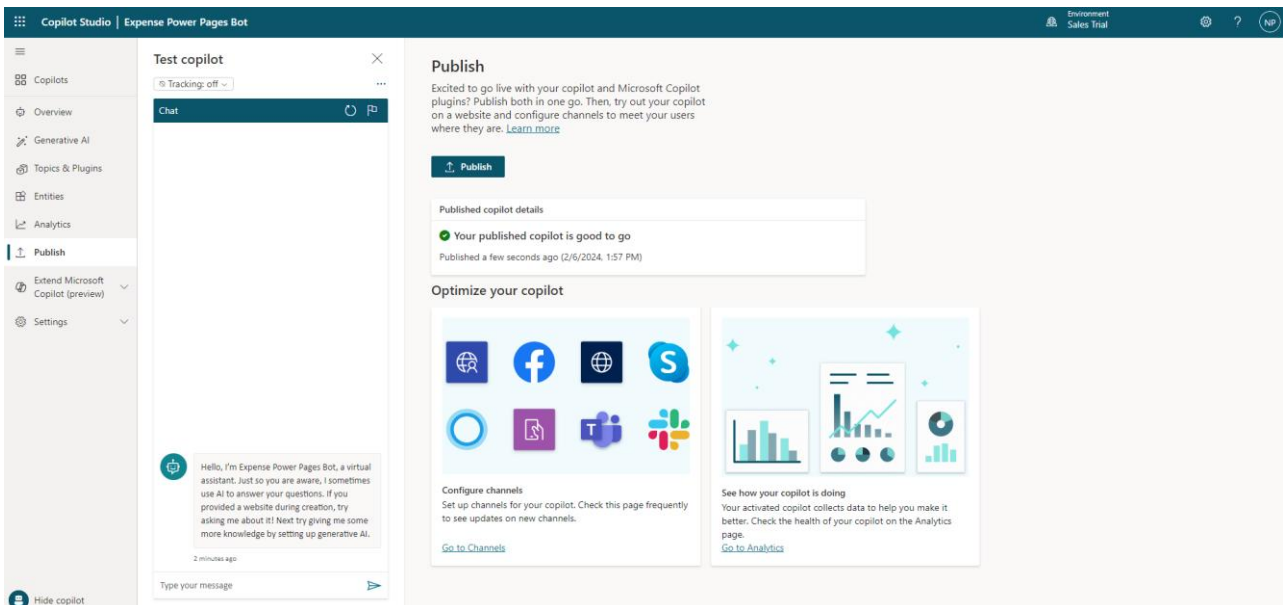
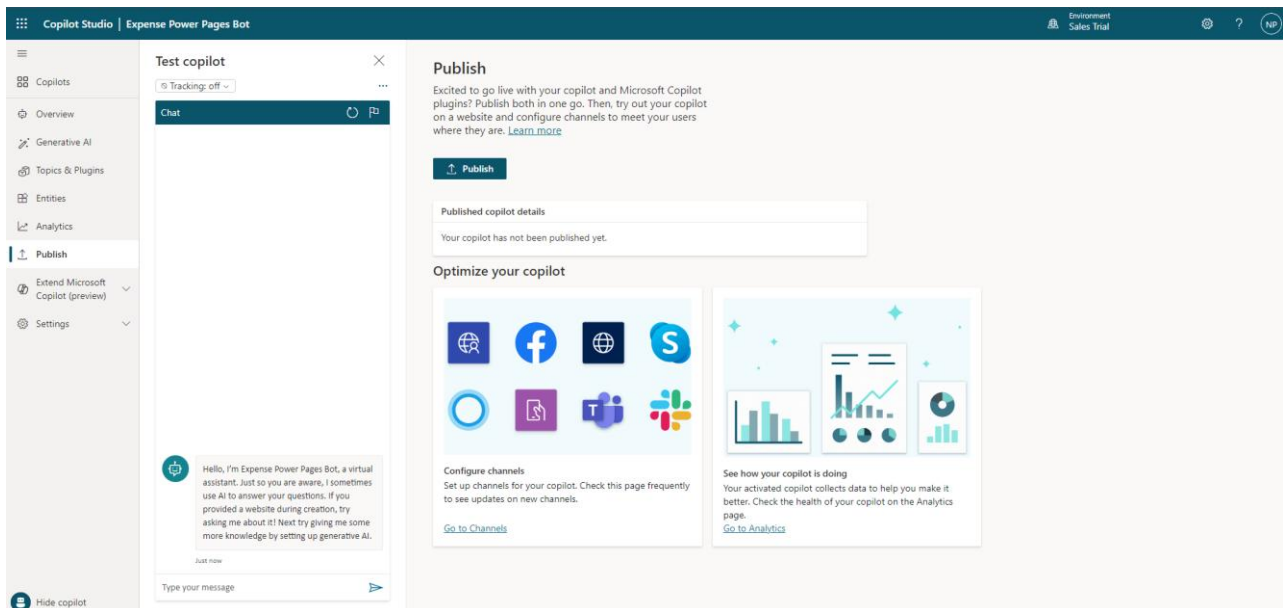
In order to read the label, we have to specify the correct format of the Status of the Expense field in our automated flow.



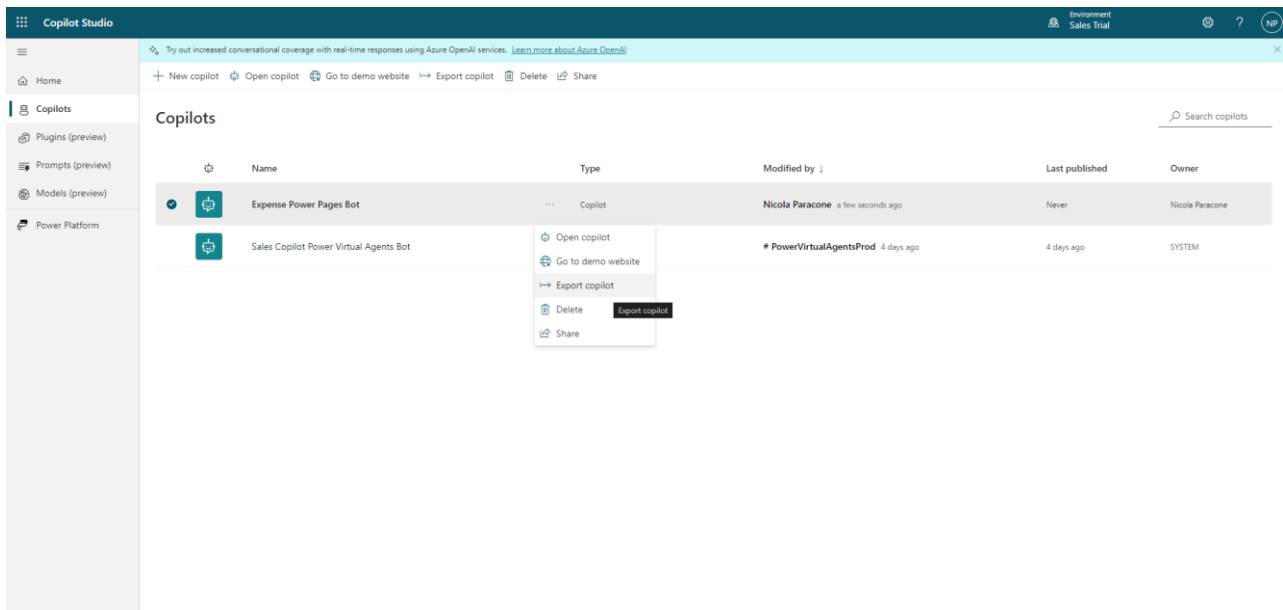
The result of asking again the chatbot for the status of a certain expense will now be the following.



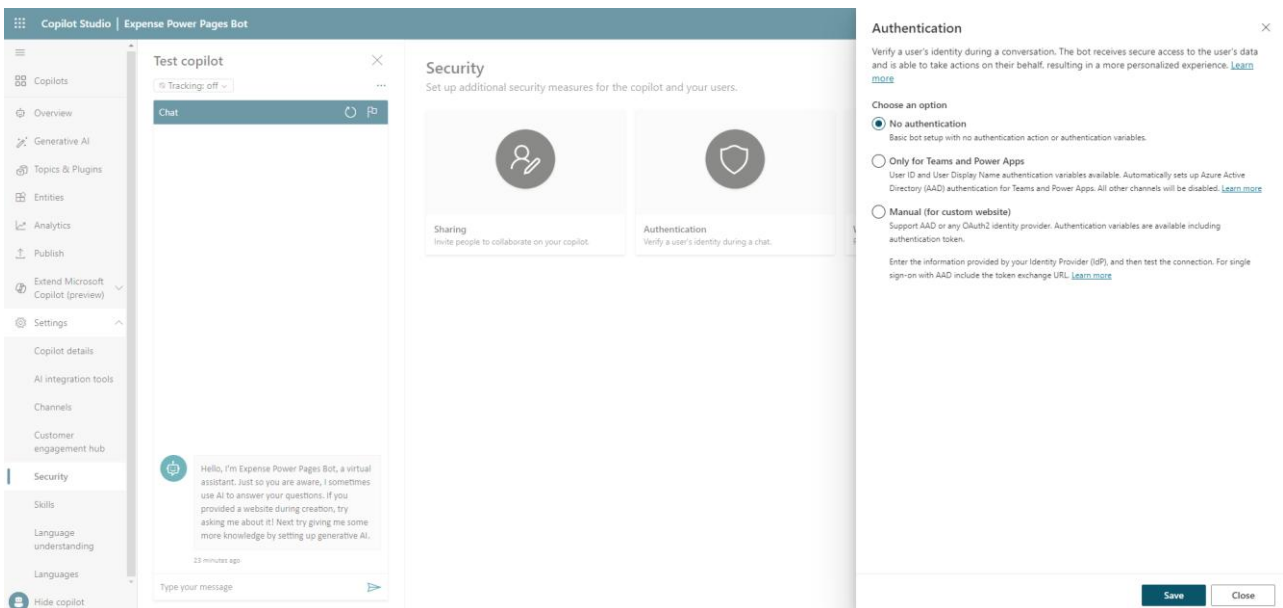
At this point, our copilot is ready to be published.



Once the copilot has been created and published, we have the possibility to export it to another environment.

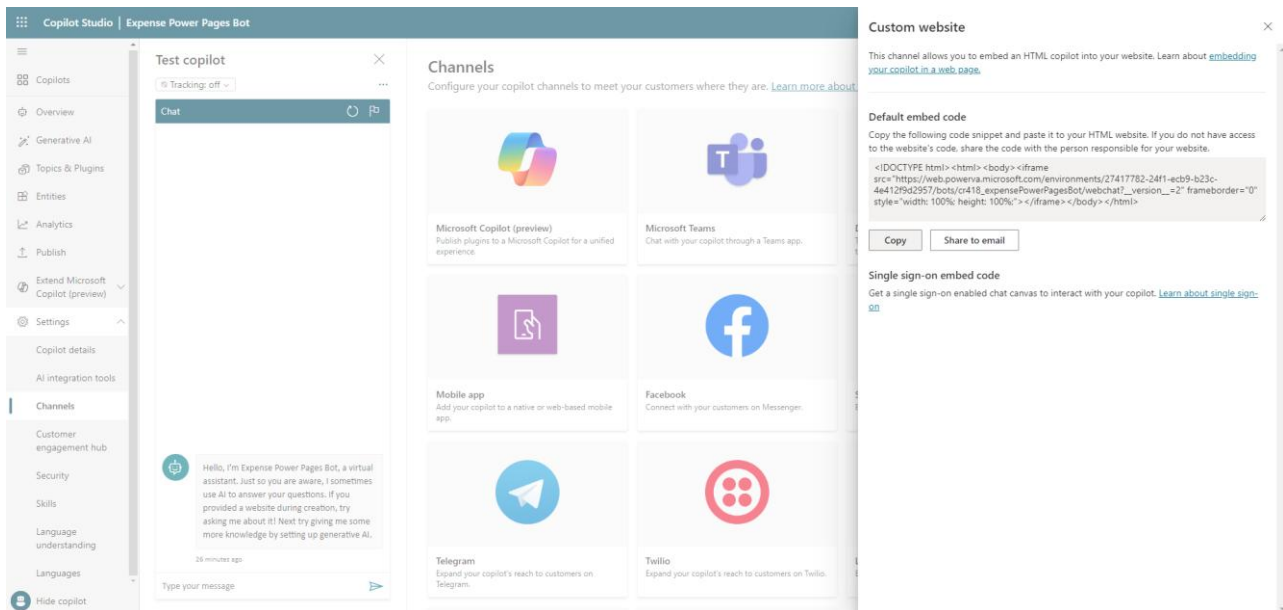


Since we now want to use our chatbot in a website, another thing we have to do is to disable authentication.

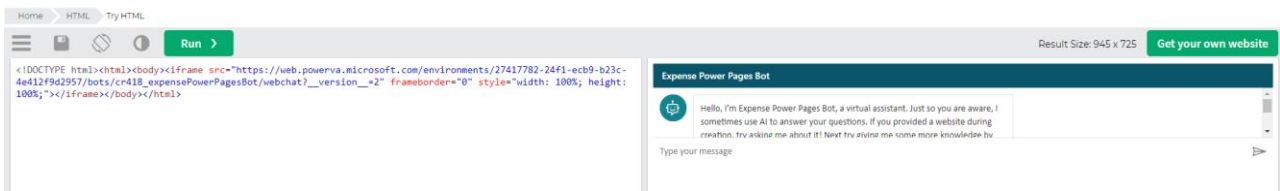


Now we go to Channels and select a channel to which add our copilot.

We click on Custom website and copy the code snippet which is provided.



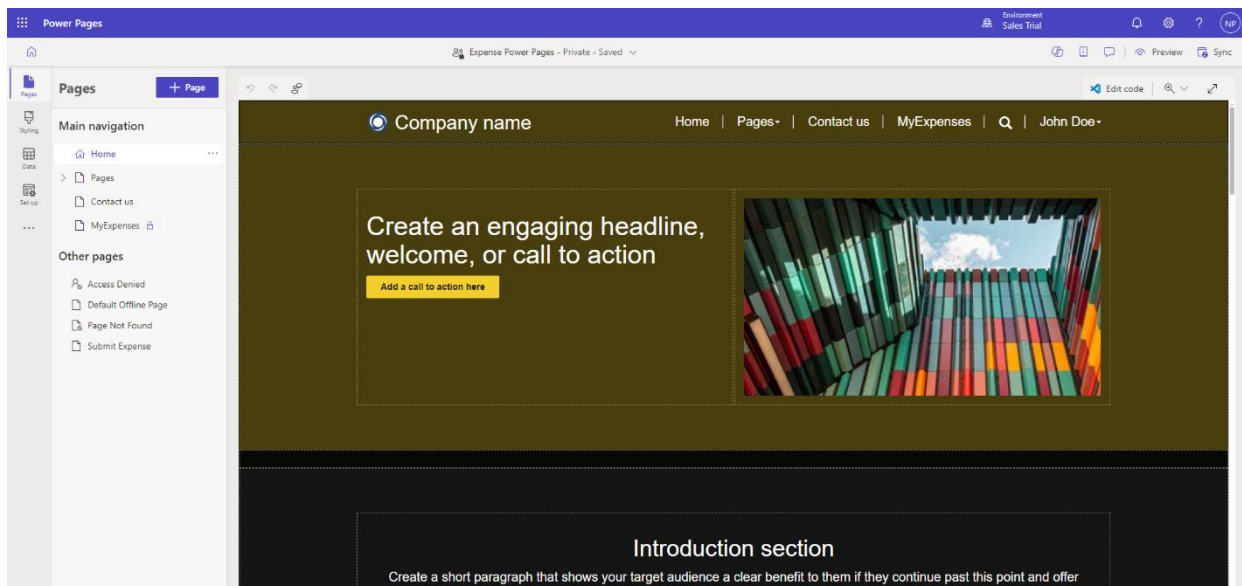
After republishing the copilot, we can navigate to https://www.w3schools.com/html/tryit.asp?filename=tryhtml_default and paste our code snippet.



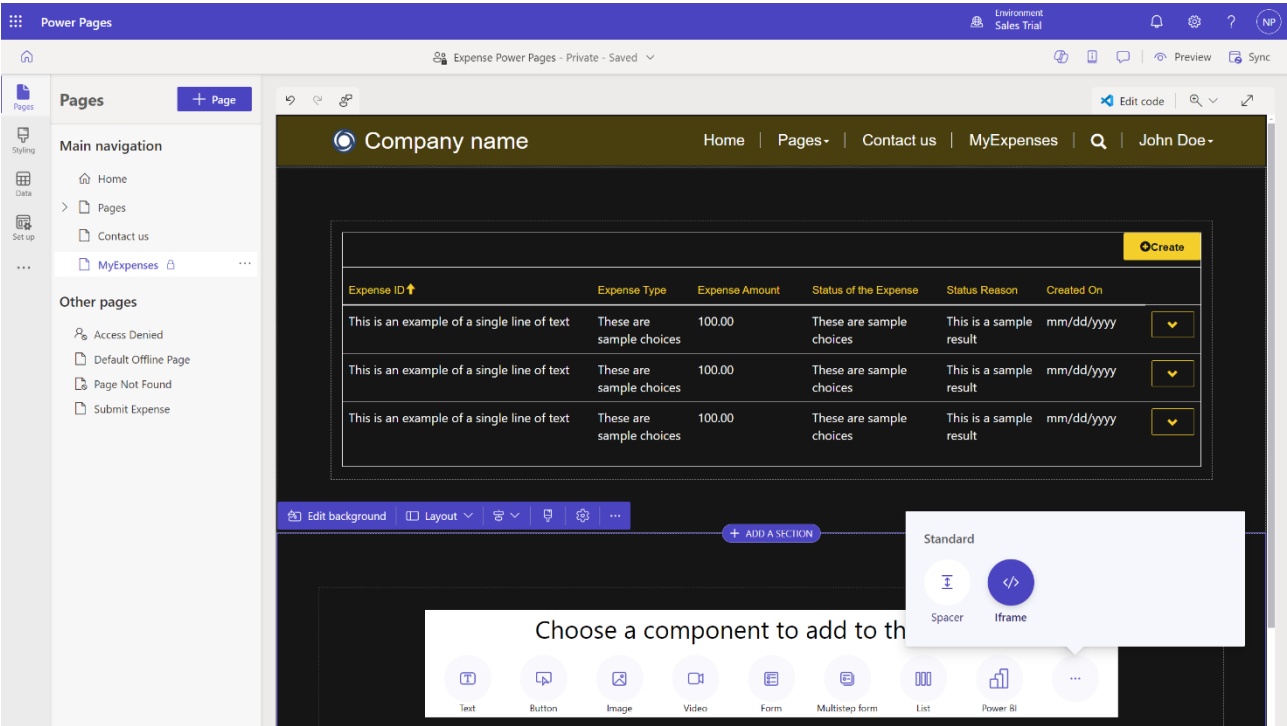
What we are seeing here is an iframe pointing to our copilot.

2. Add a chatbot to a website

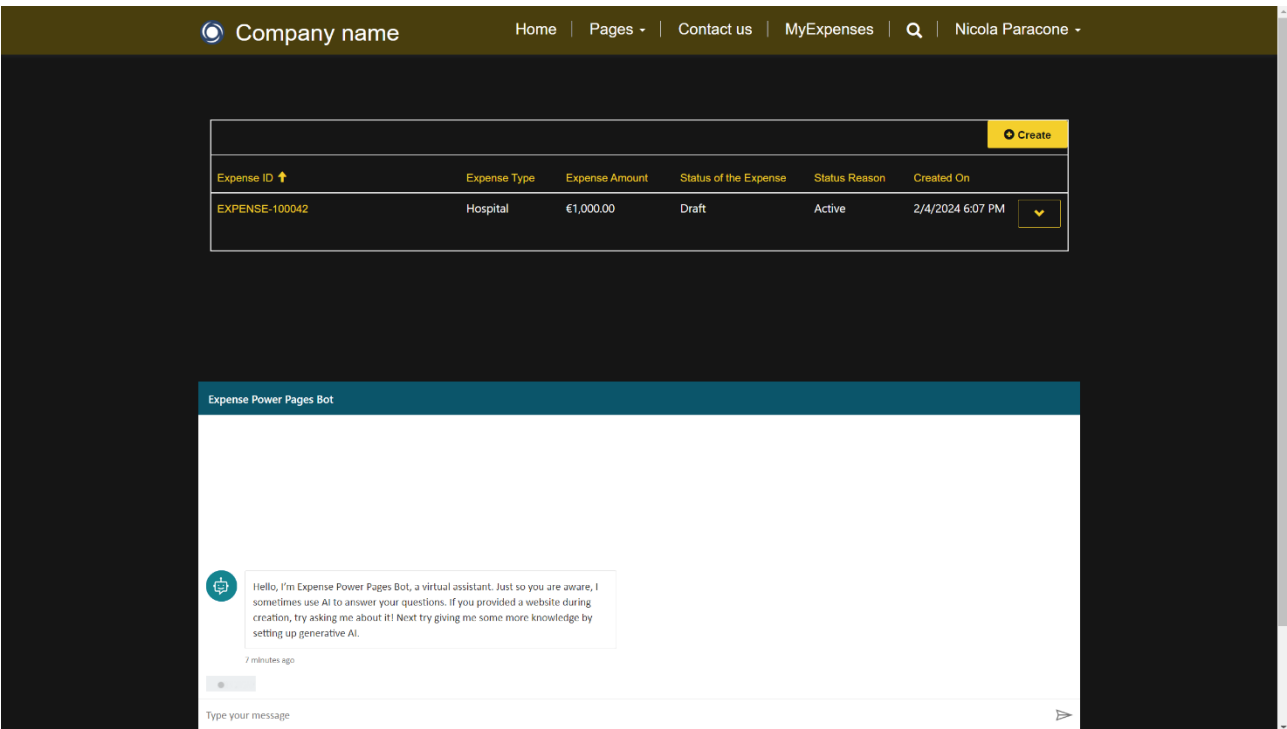
Let us open our Expense Power Pages site.



In our MyExpenses page we add a section and an iframe. For the latter, we use the URL which can be seen in the code snippet we discussed above.



In this way, our copilot will appear at the bottom of the MyExpenses page.



Notice that, in the meantime, we have also added Status of the Expense to the grid which is displayed in our site.

						Create
Expense ID ↑	Expense Type	Expense Amount	Status of the Expense	Status Reason	Created On	
EXPENSE-100042	Hospital	€1,000.00	Draft	Active	2/4/2024 6:07 PM	

Expense Power Pages Bot

setting up generative AI.

8 minutes ago

status

Just now

Can you please enter your expense Id?

Just now

EXPENSE-100042

Just now

The status of your expense is Draft.

Just now

Type your message